Coding & Solution

Date	7 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/nikhilhanumantu/Smart-Lender

Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Imbalanced-learn (SMOTE)
Key Features Implemented	Multi-model training (Decision Tree, Random Forest, KNN & XGBoost), best model selection, loan prediction, EMI calculator, risk assessment, analytics dashboard and personalized recommendations.
Pending / Incomplete Features	None
Setup / Run Instructions	Install dependencies and run app.py to launch the application.

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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
		Yes

6	Code is committed to a version control repository	

Additional Notes / Comments:

Smart Lender is an end-to-end machine learning application that preprocesses loan data, compares multiple ML models, selects the best-performing model and predicts loan eligibility through a Flask-based web application..